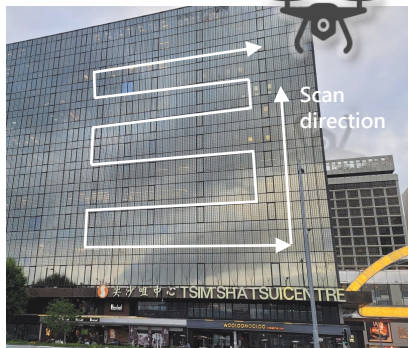
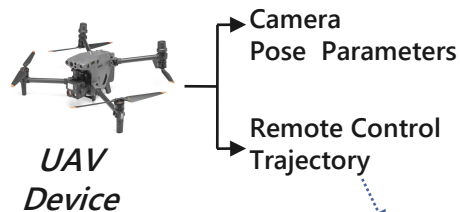
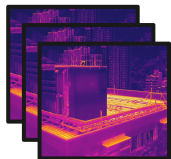




Multimodal information



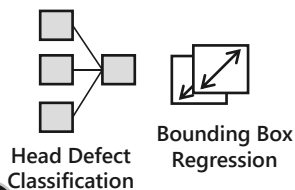
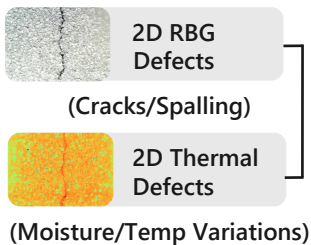
Visible Light Images



Infrared Thermal

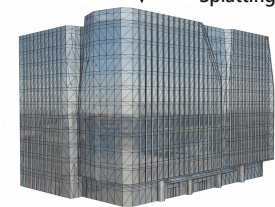
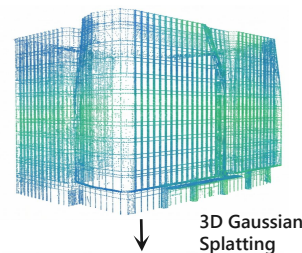


RTK Position

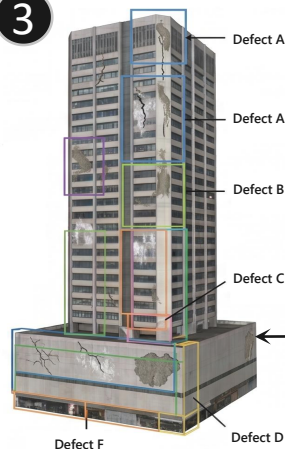


Detection Model

Cloud Point



3



Agent's Chain-of-Thought (CoT) Reasoning Steps

Coordinate(X, Y, Z, Z_{alt})
Area(S_{defect})
Temperature Diff(ΔT)

E.g. Analyze S_{defect} at Z_{alt} ...
R1: Infer cause of repair plan.
R2: Infer repair analysis on code X.
R3: Propose repair based on code X.



LMM Agent

Structured Engineering Diagnosis Report Output

- Risk Level
- Physical Cause Analysis
- Code-Compliant Repair Plan



Local Standard



RAG Retrieval

Knowledge Base